



Province of the
EASTERN CAPE
EDUCATION

Iphondo leMpuma Kapa: Isabe leMfundo
Provinsie van die Oos Kaap: Departement van Onderwys
Poratensie Ya Kapa Botjhabela: Lefapha la Thuto

**NATIONAL
SENIOR CERTIFICATE/
*NASIONALE
SENIOR SERTIFIKAAT***

GRADE/*GRAAD* 11

NOVEMBER 2025

**MATHEMATICS P2/*WISKUNDE V2*
MARKING GUIDELINE/*NASIENRIGLYN***

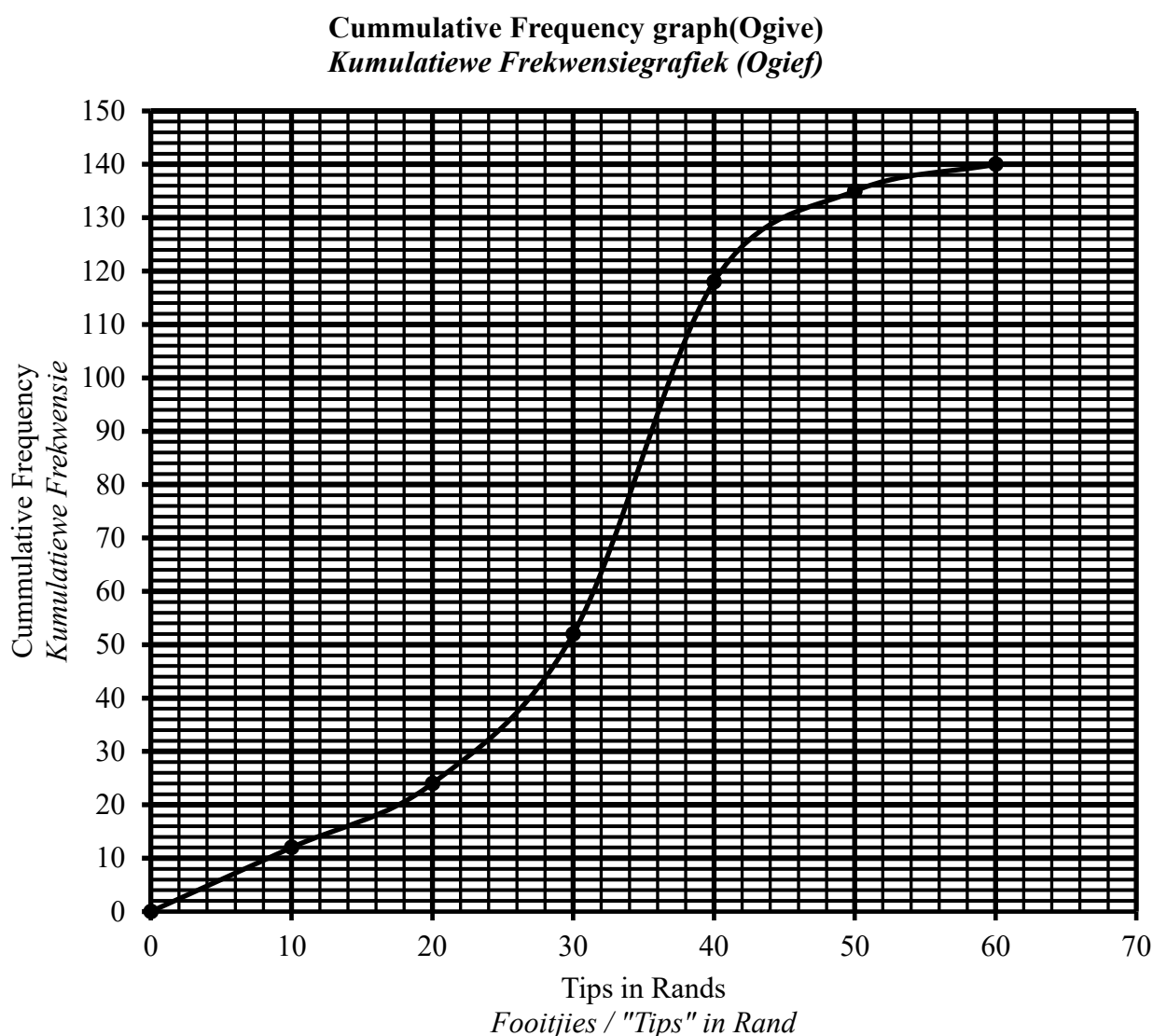
MARKS/*PUNTE*: 150

This marking guideline consists of 16 pages./
Hierdie nasienriglyn bestaan uit 16 bladsye.

QUESTION 1/VRAAG 1

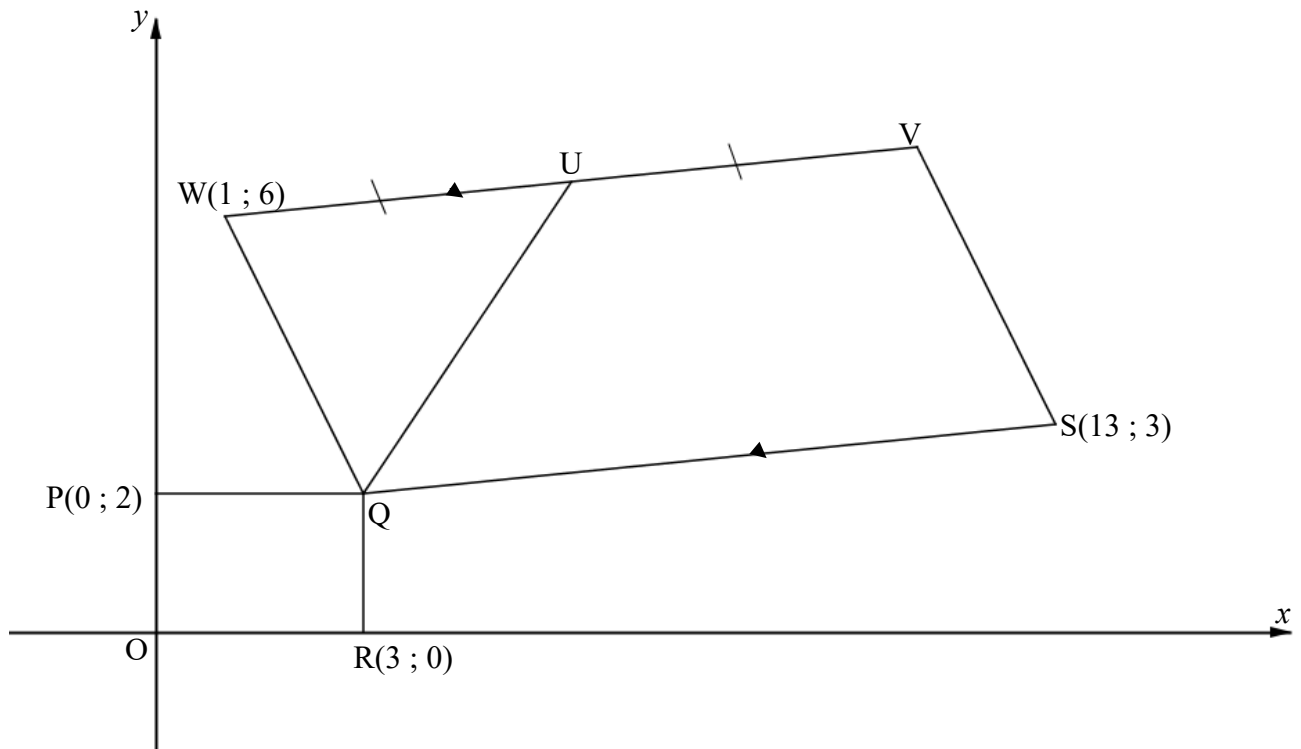
Data Set/ <i>Datastel</i> :										
Prizes/ <i>Pryse</i>	3 600	4 500	5 400	6 200	x	12 300	15 800	y	43 800	87 500
1.1	$\frac{x + 12300}{2} = 9700$ $x = 7100$					✓method/ <i>metode</i> ✓ answer/ <i>antwoord</i>				(2)
1.2	$\frac{y + 186200}{10} = 21770$ $y = 31500$					✓correct equation <i>korrekte vergelyking</i> ✓ $y + 186200$ ✓ answer/ <i>antwoord</i>				(3)
1.3	outlier/ <i>uitskieter</i> = 87 500					✓ answer/ <i>antwoord</i>				(1)
1.4	$\delta = 25258,19$					✓✓ answer/ <i>antwoord</i>				(2)
1.5	Mean will increase by R200 <i>Gemiddelde sal vermeerder met R200</i> Median will increase by R200 <i>Mediaan sal vermeerder met R200</i> Standard deviation will remain the same <i>Standaardafwyking sal dieselfde bly</i>					✓ increase by R200 <i>vermeerder met R200</i> ✓ increase by R200 <i>vermeerder met R200</i> ✓ remain the same <i>bly dieselfde</i>				(3)
										[11]

QUESTION 2/VRAAG 2



2.1	140 employees/werknemers	✓ answer/antwoord	(1)
2.2	$30 < x \leq 40$	✓ answer/antwoord	(1)
2.3	Reading from the graph / lees vanaf grafiek 32 employees/werknemers	✓ Reading/lees ✓ answer/antwoord	(2)
2.4	$118 - 24 = 94$	✓ $118 - 24$ ✓ answer/antwoord	(2)
2.5	25 th percentile is at 35 = R25 25 ^{ste} persentiel is by 35 = R25 75 th percentile is at 105 = R38 75 ^{ste} persentiel is by 105 = R38	✓ Q_1 ✓ Q_3	(2)
2.6	$IQR/IKW = Q_3 - Q_1 = R38 - R25$ $= R13$	✓ method/metode ✓ answer/antwoord	(2)
			[10]

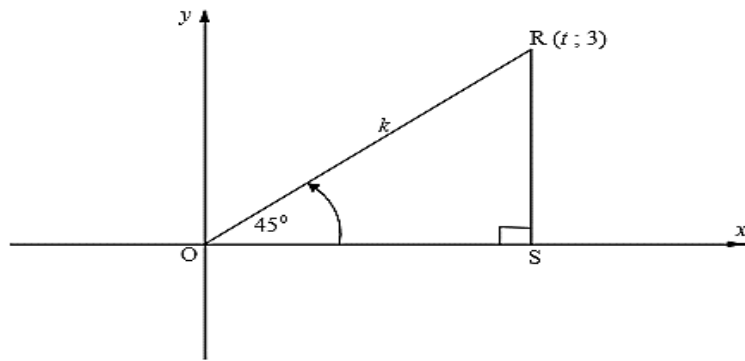
QUESTION 3/VRAAG 3



3.1	$Q(3;2)$	✓ answer/antwoord	(1)
3.2	$m_{QS} = \frac{3-2}{13-3}$ $= \frac{1}{10}$	✓ correct substitution korrekte vervanging ✓ answer/antwoord	(2)
3.3	$m_{VW} = m_{QS} = \frac{1}{10}$ $6 = \frac{1}{10}(1) + c$ $c = \frac{59}{10}$ $y = \frac{1}{10}x + \frac{59}{10}$ OR/OF $y - 6 = \frac{1}{10}(x - 1)$ $y = \frac{1}{10}x + \frac{59}{10}$	✓ gradient of VW gradiënt van VW ✓ substitution/vervanging ✓ answer/antwoord	(3)
3.4	$\frac{3}{2}x - \frac{5}{2} = \frac{1}{10}x + \frac{59}{10}$ $\frac{7}{5}x = \frac{42}{5}$ $x_U = 6$ $y_U = \frac{3}{2}(6) - \frac{5}{2} = \frac{13}{2}$ $6 = \frac{1+x_V}{2}; \frac{13}{2} = \frac{6+y_V}{2}$ $11 = x_V; 5 = y_V$	✓ equating equations gelykstel van vergelykings ✓ simplification vereenvoudiging ✓ x-value / x-waarde ✓ y-value / y-waarde ✓ method/metode	(5)

3.5	$M_{WS}\left(\frac{13+1}{2}; \frac{3+6}{2}\right)$ $M_{WS}\left(7; \frac{9}{2}\right)$ $M_{QV}\left(\frac{11+3}{2}; \frac{7+2}{2}\right)$ $M_{QV}\left(7; \frac{9}{2}\right)$ $\therefore$ WVSQ is a parm (Diag. bisect each other.) WVSQ is 'n $\parallel^m$ (Hoeklyne halveer mekaar)	✓ correct substitution korrekte vervanging ✓ coordinates of midpt of WS koördinate van midpt van WS ✓ coordinates of midpt of QV koördinate van midpt van QV ✓ reason/rede	(4)
3.6	$WV = \sqrt{(1-11)^2 + (6-7)^2}$ $= \sqrt{101}$	✓ correct substitution korrekte vervanging ✓ answer/antwoord	(2)
3.7	Let inclination of WV to be θ and QU be α Laat inklinasie van WV, θ wees en van QU, α wees $\tan \theta = m_{WV} = \frac{1}{10}$ $\theta = 5,71^\circ$ $\tan \alpha = m_{QU} = \frac{3}{2}$ $\alpha = 56,31^\circ$ $\therefore \hat{WUQ} = \hat{UQS} = 50,6^\circ$ [alt $\angle$s, $WV \parallel QS$] [verv. $\angle$e, $WV \parallel QS$]	✓ $\tan \theta = m_{WV}$ ✓ size of θ / grootte van θ ✓ size of α / grootte van α ✓ answer/antwoord	(4)
3.8	$WU = \frac{\sqrt{101}}{2}$ $UQ = \sqrt{(6-3)^2 + \left(\frac{13}{2} - 2\right)^2} = \frac{3\sqrt{13}}{2}$ $A. \text{ of } \triangle QUW = \frac{1}{2} \times \frac{3\sqrt{13}}{2} \times \frac{\sqrt{101}}{2} \times \sin(50,60^\circ)$ $= 10,50$ $\frac{\text{Area of PQRO}}{\text{Area of } \triangle QUW} = \frac{2 \times 3}{10,50}$ $= \frac{4}{7}$	✓ WU ✓ UQ ✓ correct subst. in to Area rule korrekte verv. in oppervl. reël ✓ A. of Δ. / Oppervl. van Δ ✓ correct subst. in A. of rect. korrekte verv. in Opp. van regh ✓ answer/antwoord	(6)
			[27]

QUESTION 4 / VRAAG 4

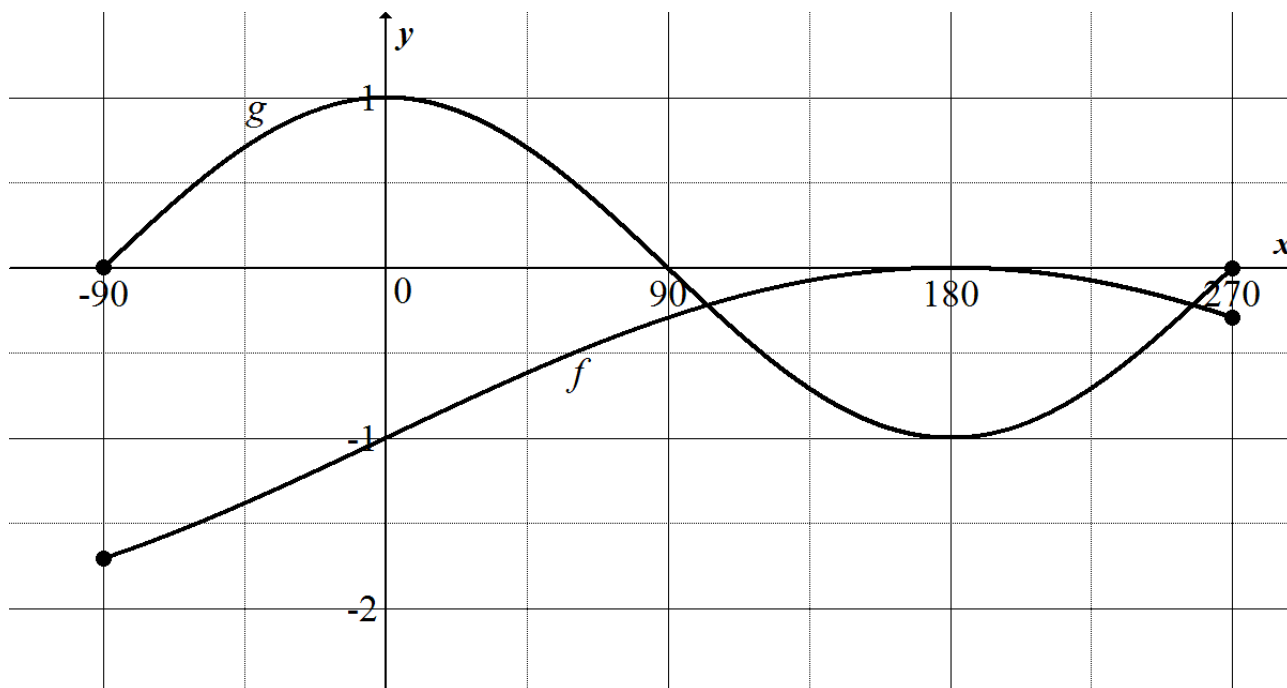


4.1.1	$\tan 45^\circ = \frac{3}{t}$ $t = 3$	✓ correct trig ratio ✓ korrekte trig. verhouding ✓ t-value/waarde	(2)
4.1.2	$\sin 45^\circ = \frac{3}{k}$ OR / OF $\cos 45^\circ = \frac{t}{k} = \frac{3}{k}$ $k = 3\sqrt{2}$ $k = 3\sqrt{2}$	✓ correct trig ratio ✓ korrekte trig. verhouding ✓ k-value/waarde	(2)
4.2.1	$8\cos^2 \alpha - 2 = 0$ $\cos^2 \alpha = \frac{1}{4}$ $\cos \alpha = \frac{1}{2}$	✓ simplification ✓ vereenvoudiging ✓ answer/antwoord	(2)
4.2.2	 OR / OF $y = \sqrt{2^2 - 1^2}$ <i>Pyth</i> $= \sqrt{3}$ $4\left[\tan^2 \alpha - \cos^2 \alpha\right] = 4\left[\left(\frac{\sqrt{3}}{1}\right)^2 - \left(\frac{1}{2}\right)^2\right]$ $= 11$	✓ diagram/diagram OR / OF ✓ $y = \sqrt{3}$ ✓ correct substitution ✓ korrekte vervanging ✓ answer/antwoord	(3)

4.3.1	$\frac{\cos x \cdot \tan x}{2[\sin x \cdot \cos(x - 90^\circ) + \cos x \cdot \cos(-x)]}$ $= \frac{\cos x \cdot \frac{\sin x}{\cos x}}{2[\sin x \cdot (\sin x) + \cos x \cdot (\cos x)]}$ $= \frac{\sin x}{2[\sin^2 x + \cos^2 x]}$ $= \frac{\sin x}{2(1)}$ $= \frac{\sin x}{2}$	$\checkmark \frac{\sin x}{\cos x}$ $\checkmark \sin x \quad \checkmark \cos x$ $\checkmark$ simplification <i>vereenvoudiging</i> $\checkmark \sin^2 x + \cos^2 x = 1$ $\checkmark$ answer/antwoord	(6)
4.3.2	Period/Periode = 360°	$\checkmark$ answer/antwoord	(1)
4.4	$\frac{\cos 42^\circ \sin 48^\circ - \tan^2(-45^\circ)}{\cos^2 132^\circ}$ $= \frac{\cos 42^\circ \cdot \cos 42^\circ - (-\tan 45^\circ)^2}{(\cos(90^\circ + 42^\circ))^2}$ $= \frac{\cos^2 42^\circ - (-1)^2}{(-\sin 42^\circ)^2}$ $= \frac{\cos^2 42^\circ - 1}{\sin^2 42^\circ}$ $= \frac{1 - \sin^2 42^\circ - 1}{\sin^2 42^\circ}$ $= -1$	$\checkmark \sin 48^\circ = \cos 42^\circ$ $\checkmark (-\tan 45^\circ)^2$ $\checkmark (-\sin 42^\circ)^2$ $\checkmark \cos^2 42^\circ = 1 - \sin^2 42^\circ$ $\checkmark$ answer/antwoord	(5)
4.5.1	$\text{LHS/LK} = \tan \beta - \sin \beta \cdot \cos \beta$ $= \frac{\sin \beta}{\cos \beta} - \sin \beta \cdot \cos \beta$ $= \frac{\sin \beta - \sin \beta \cos^2 \beta}{\cos \beta}$ $= \frac{\sin \beta (1 - \cos^2 \beta)}{\cos \beta}$ $= \tan \beta \cdot \sin^2 \beta$ $= \text{RHS / RK}$	$\checkmark \frac{\sin \beta}{\cos \beta}$ $\checkmark$ simplification <i>vereenvoudiging</i> $\checkmark$ common factor <i>gemeenskaplike faktor</i>	(3)
4.5.2	$\beta = -90^\circ$ and / en $\beta = 90^\circ$	$\checkmark \beta = -90^\circ \quad \checkmark \beta = 90^\circ$	(2)

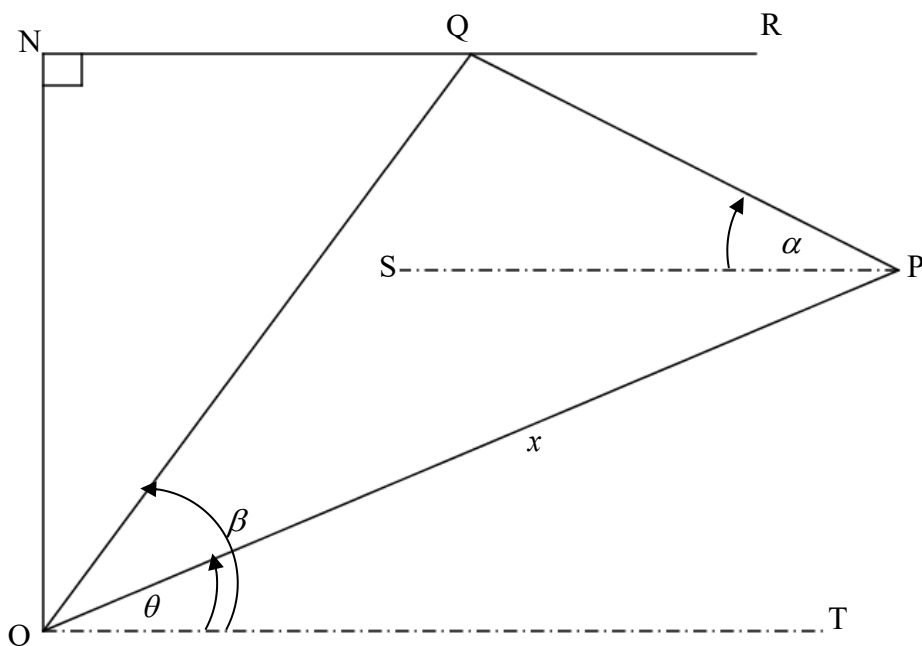
4.6	$3 \sin \theta = -2 \cos^2 \theta$ $2(1 - \sin^2 \theta) + 3 \sin \theta = 0$ $2 - 2 \sin^2 \theta + 3 \sin \theta = 0$ $2 \sin^2 \theta - 3 \sin \theta - 2 = 0$ $(2 \sin \theta + 1)(\sin \theta - 2) = 0$ $\sin \theta = -\frac{1}{2}$ or / of $\sin \theta \neq 2$ $\theta = 210^\circ + 360.k \quad \theta = 330^\circ + 360.k, \quad k \in \mathbb{Z}$	$\checkmark 1 - \sin^2 \theta$ $\checkmark$ standard form <i>standaardvorm</i> $\checkmark$ factors/faktore $\checkmark$ both equations <i>beide vergelykings</i> $\checkmark \theta = 210^\circ + 360.k$ $\checkmark \theta = 330^\circ + 360.k, \quad k \in \mathbb{Z}$	(6)
			[32]

QUESTION 5/VRAAG 5



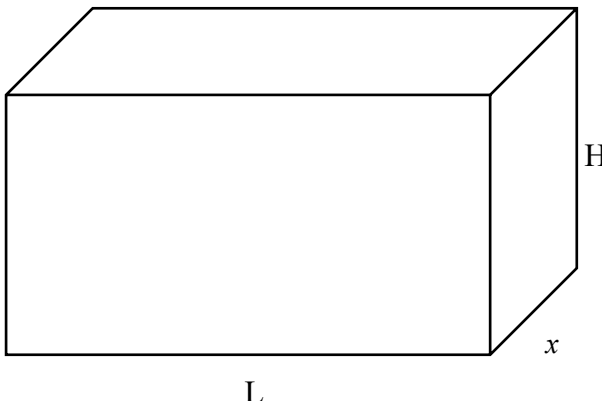
5.1	$a = 1$ $q = 0$	✓ value of a / waarde van a ✓ value of q / waarde van q	(2)
5.2	$y \in [-2; 0]$ or / of $-2 \leq y \leq 0$	✓ correct critical value korrekte kritieke waarde ✓ correct notation korrekte notasie	(2)
5.3	Amplitude = 1	✓ correct answer korrekte antwoord	(1)
5.4	$x = 0^\circ$	✓ answer / antwoord	(1)
5.5	$-90^\circ \leq x \leq 90^\circ$	✓✓ correct answer korrekte antwoord	(2)
5.6	$h(x) = g(x + 90^\circ)$ $= \cos(x + 90^\circ)$ $= -\sin x$	✓ $\cos(x + 90^\circ)$ ✓ $-\sin x$	(2)
			[10]

QUESTION 6/VRAAG 6

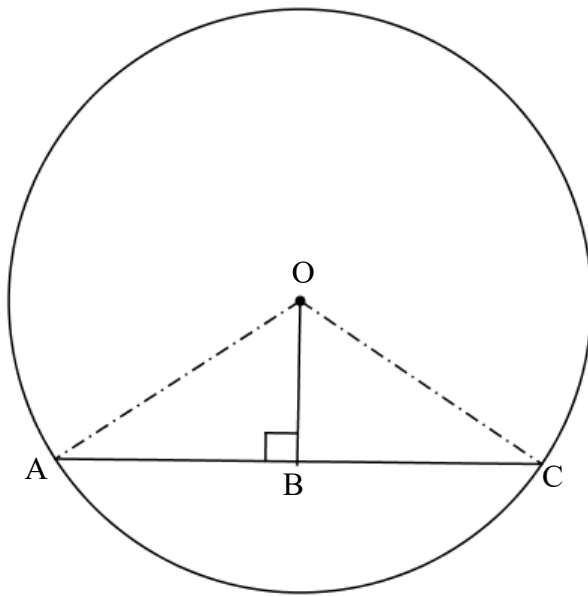


6.1	$\hat{SPO} = \theta + \alpha$ [alt. $\angle$ s, verw. $\angle$ e : $SP \parallel OT$] $\therefore$ $\hat{QOP} = \beta - \theta$ $\hat{OQP} = 180^\circ - (\theta + \alpha) - (\beta - \theta)$ [$\angle$ s in Δ / $\angle$ e in Δ] $= 180^\circ - (\beta + \alpha)$	$\checkmark \hat{SPO} = \theta + \alpha$ $\checkmark$ using sum of angles in Δ <i>gebruik van som van $\angle$e in Δ</i>	(2)
6.2	$\frac{OQ}{\sin \hat{P}} = \frac{OP}{\sin \hat{Q}}$ $\frac{OQ}{\sin(\alpha + \theta)} = \frac{x}{\sin[180^\circ - (\beta + \alpha)]}$ $OQ = \frac{x \cdot \sin(\alpha + \theta)}{\sin(\beta + \alpha)}$	$\checkmark$ application of sine rule <i>toepassing van sinusreël</i> $\checkmark$ correct substitution <i>korrekte vervanging</i> $\checkmark$ reduction $\sin(\beta + \alpha)$ <i>reduksie $\sin(\beta + \alpha)$</i>	(3)
6.3	$\frac{5\sqrt{2}}{\sin 45^\circ} = \frac{x}{\sin 90^\circ}$ $x = 10$	$\checkmark$ correct substitution <i>korrekte vervanging</i> $\checkmark$ answer/antwoord	(2)
			[7]

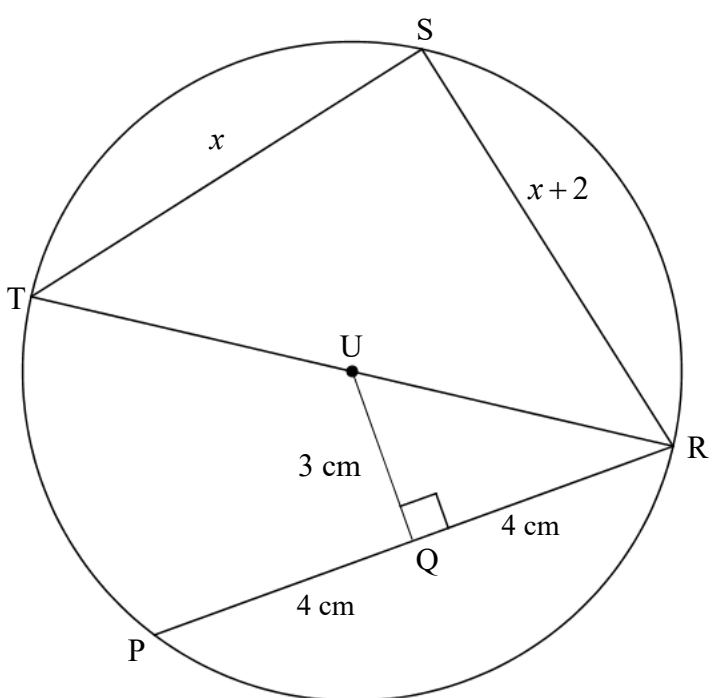
QUESTION 7/VRAAG 7

			
7.1	$L = 2x$ $H = 3 + x$	✓ $L = 2x$ ✓ $H = 3 + x$	(2)
7.2	$SA = 2(2x \times x) + 2(x(3 + x)) + 2(2x(3 + x))$ $= 4x^2 + 6x + 2x^2 + 12x + 4x^2$ $= 10x^2 + 18x$	✓ correct substitution / korrekte vervanging ✓ simplification / vereenvoudiging ✓ answer / antwoord	(3)
7.3	$10x^2 + 18x = 784$ $10x^2 + 18x - 784 = 0$ $5x^2 + 9x + 392 = 0$ $(5x + 49)(x - 8) = 0$ $x \neq \frac{40}{5}$ or $x = 8$	✓ correct substitution/ korrekte vervanging ✓ standard form/standaardvorm ✓ factors/faktore ✓ correct answer/korrekte antwoord	(4)
7.4	$V = (8)(2 \times 8)(3 + 8)$ $= 1408$	✓ correct substitution into volume / korrekte vervanging in volume ✓ answer/antwoord	(2)
			[11]

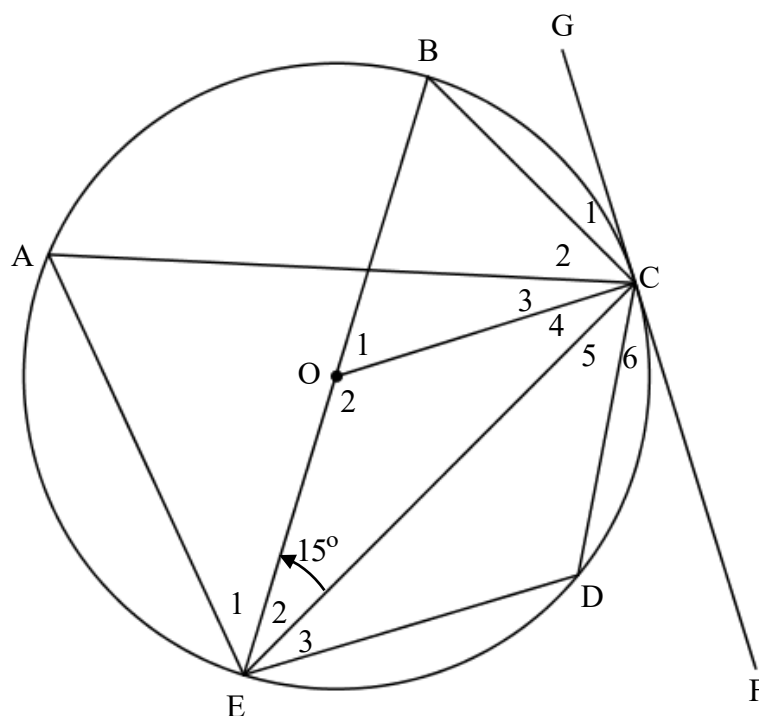
QUESTION 8/VRAAG 8



8.1	Constructions: Join OA and OC <i>Konstruksies: Verbind OA en OC</i> Proof in $\triangle AOB$ and $\triangle COB$ <i>Bewys in $\triangle AOB$ en $\triangle COB$</i> 1. $AO = CO$ [radii]/[radiusse] 2. $\angle ABO = \angle CBO = 90^\circ$ [$\angle$s on a str. line]/[$\angle$op reguitlyn] 3. $OB = OB$ [common]/[gemeen] $\therefore \triangle AOB \equiv \triangle COB$ [RHS]/[90° Sk S] $\therefore AB = BC$	✓ construction <i>konstruksie</i> ✓ S/R ✓ S/R ✓ S ✓ R	(5)
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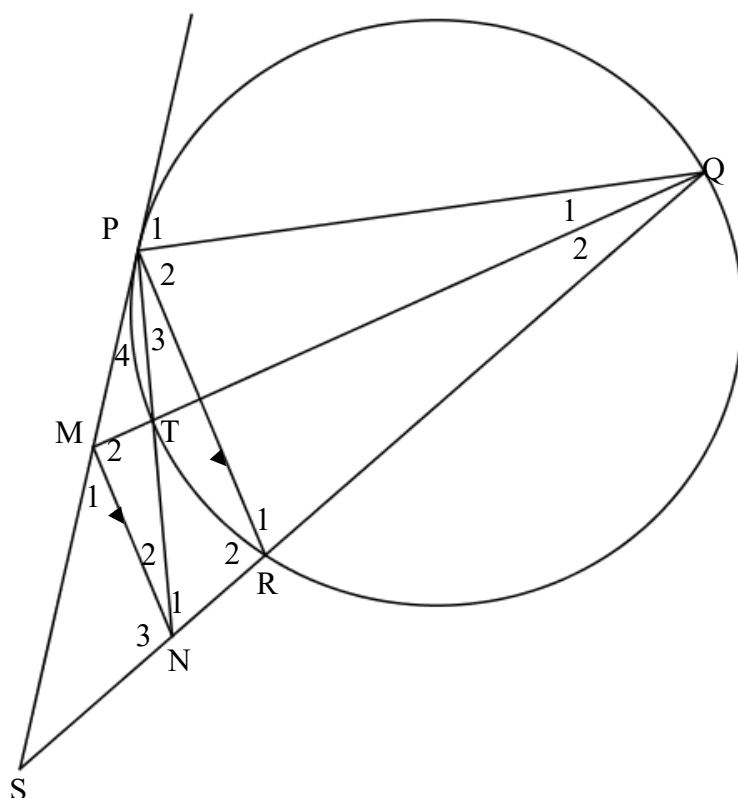
8.2			
8.2.1	$PQ = QR = 4 \text{ cm}$ [line from centre $\perp$ to chord] [lyn vanaf middelpunt $\perp$ op koord] $UR = \sqrt{4^2 + 3^2}$ [Pyth.theo]/ [Stelling van Pythagoras] $= 5$	$\checkmark$ S $\checkmark$ R $\checkmark$ correct substitution/ korrekte vervanging $\checkmark$ answer/antwoord	(4)
8.2.2	$TR = 2(UR) = 10$ [diameter]/ [middellyn] $\hat{S} = 90^\circ$ [$\angle$ in semi – circle]/ [$\angle$ in semi – sirkel] $10^2 = x^2 + (x + 2)^2$ [Pyth. theo]/ [Stelling van Pythagoras] $100 = x^2 + x^2 + 4x + 4$ $2x^2 + 4x - 96 = 0$ $x^2 + 2x - 48 = 0$ $(x - 6)(x + 8) = 0$ $x = 6 \text{ or / of } x \neq -8$ $\therefore TS = 6$	$\checkmark$ S $\checkmark$ R $\checkmark$ correct substitution/ korrekte vervanging $\checkmark$ standard form / standaardvorm $\checkmark$ factors / faktore $\checkmark$ correct value of x / korrekte waarde van x	(6)
			[15]

QUESTION 9 / VRAAG 9



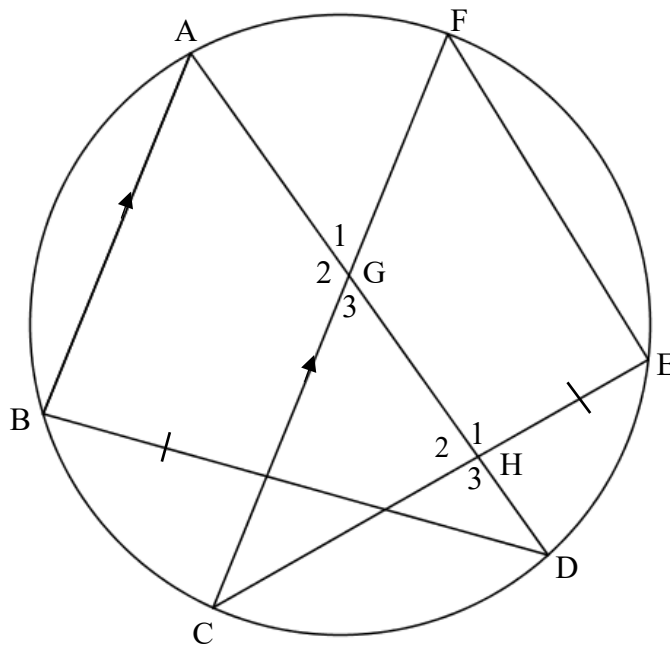
9.1.1	$\hat{O}_1 = 30^\circ$ [$\angle$ at centre = $2\angle$ at circumf.] [Middelpunts $\angle = 2 \times$ Omtreks $\angle$]	✓ S ✓ R	(2)
9.1.2	$\hat{O}_2 = 150^\circ$ $\hat{A} = 75^\circ$ [$\angle$ s on a str.line]/[$\angle$ op reguitlyn] [$\angle$ at centre = $2\angle$ at circumf.] [Middelpunts $\angle = 2 \times$ Omtreks $\angle$]	✓ S ✓ S/R	(2)
9.1.3	$\hat{D} = 105^\circ$ [opp. $\angle$ s of a cyclic quad] [Teenoorst. $\angle$ e van koordevierhoek]	✓ S ✓ R	(2)
9.1.4	$\widehat{ECF} = 75^\circ$ [tan chord theo.]/[raaklyn-koord stelling]	✓ S ✓ R	(2)
9.1.5	$\widehat{OCG} = 90^\circ$ [rad $\perp$ tan]/[radius $\perp$ raaklyn]	✓ S ✓ R	(2)

9.2



9.2.1	$\hat{P}_1 = \hat{R}_1$ [tan chord theo] / [raaklyn – koord stelling] $\hat{P}_1 = \hat{M}\hat{N}Q$ [corresp. $\angle$ s, / ooreenk. $\angle$ e : $MN \parallel PR$] $\hat{R}_1 = \hat{P}\hat{T}Q$ [$\angle$ s same seg] / [$\angle$ e in dies. segment]	$\checkmark$ S $\checkmark$ R $\checkmark$ S $\checkmark$ R Any 2 angles with reasons / Enige 2 hoeke met redes	(4)
9.2.2	$\hat{S}\hat{P}Q = \hat{Q}$ [tan chord theo] / [raaklyn – koord stelling] $\hat{S}\hat{P}Q = \hat{M}_1$ [corresp. $\angle$ s, / ooreenk. $\angle$ e : $MN \parallel PR$] $\therefore \hat{Q}_1 = \hat{M}_1$ $\therefore MNQP$ is a cyclic quad [conv. ext. $\angle$ of a cyclic quad] <i>MNQP is 'n koordevh. [omgek. buite $\angle$ van koordevh]</i>	$\checkmark$ S $\checkmark$ R $\checkmark$ S $\checkmark$ R	(4)
9.2.3	$\hat{P}_3 = \hat{N}_2$ [alt. $\angle$ s, $MN \parallel PR$] / [verw. $\angle$ e : $MN \parallel PR$] $\hat{N}_2 = \hat{Q}_1$ [$\angle$ s same seg] / [$\angle$ e in dies. segment] $\hat{Q}_1 = \hat{P}_4$ [tan chord theo] / [raaklyn – koord stelling] $\therefore \hat{P}_3 = \hat{P}_4$ <i>PN bisects $\hat{S}\hat{P}R$ / PN halveer $\hat{S}\hat{P}R$</i>	$\checkmark$ S/R $\checkmark$ S $\checkmark$ R $\checkmark$ S/R	(4)
			[22]

QUESTION 10/VRAAG 10



$\hat{G}_3 = \hat{A}$	[corresp. $\angle$ s / ooreenk. $\angle$ e, $GC \parallel AB$]	✓ S	✓ R	
$\hat{A} = \hat{F}$	[$\angle$ s subt by = chords] / [$\angle$ e onderspan deur = koorde]	✓ S	✓ R	
$\therefore \hat{G}_3 = \hat{F}$		✓ R		
$\therefore AB \parallel FE$	[corresp. $\angle$ s, =] / [ooreenkomstige $\angle$ e =]			[5]

TOTAL/TOTAAL: 150